

Lecture 6

Insertion Operation of Linked List

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Contents

Insertion methods at any position

Algorithm of Insertion

Inserting a new node

Possible cases of Insert Node

1. Insert into an empty list
2. Insert in front
3. Insert at back
4. Insert in middle

But, in fact, only need to handle two cases:

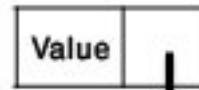
1. Insert as the first node (Case 1 and Case 2)
2. Insert in the middle or at the end of the list (Case 3 and Case 4)

Inserting at Beginning

Head



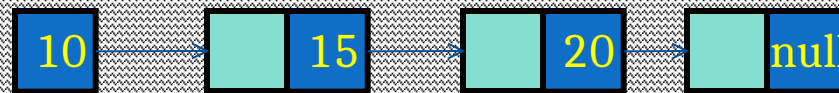
newNode 1



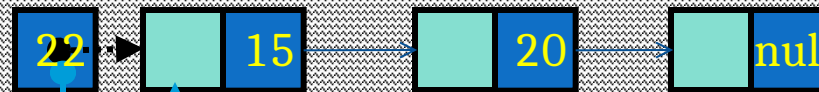
NULL

NULL

Start



START



NEW

Inserting at Beginning

Step 1. OVERFLOW, IF(AVAIL = NULL)
write : OVERFLOW and EXIT.

Step 2. Create a new node NODE=AVAIL
and assign the address to any variable say AVAIL_PTR .

Step 3. ASSIGN NODE[DATA] = ITEM

Step 4. IF(START = NULL)
ASSIGN NODE[LINK] = NULL
ELSE
ASSIGN NODE[LINK] = START

Step 5. ASSIGN START = AVAIL_PTR

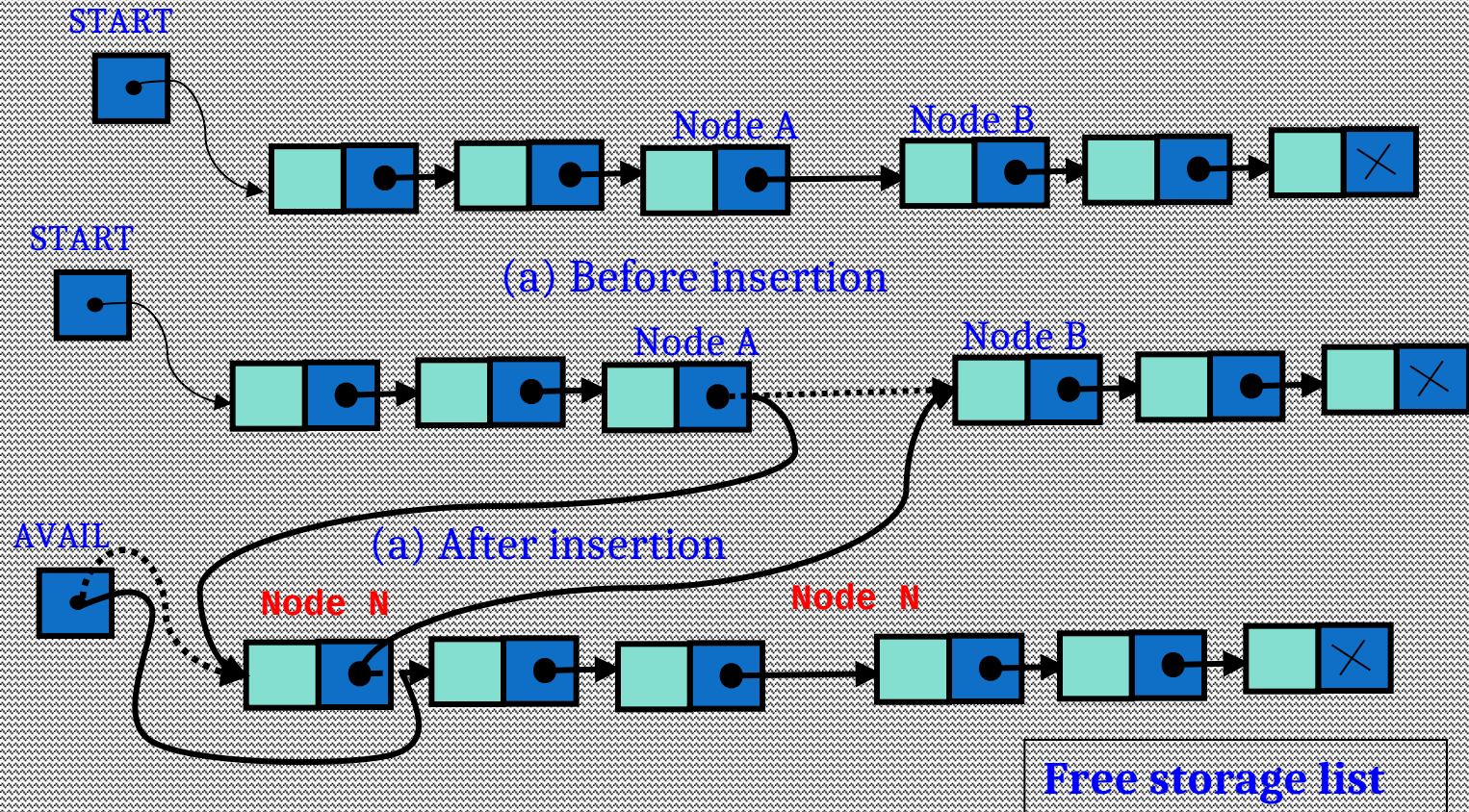
Step 6. EXIT

Insertion in the Middle

Node N is to be inserted in to the list between nodes A and B

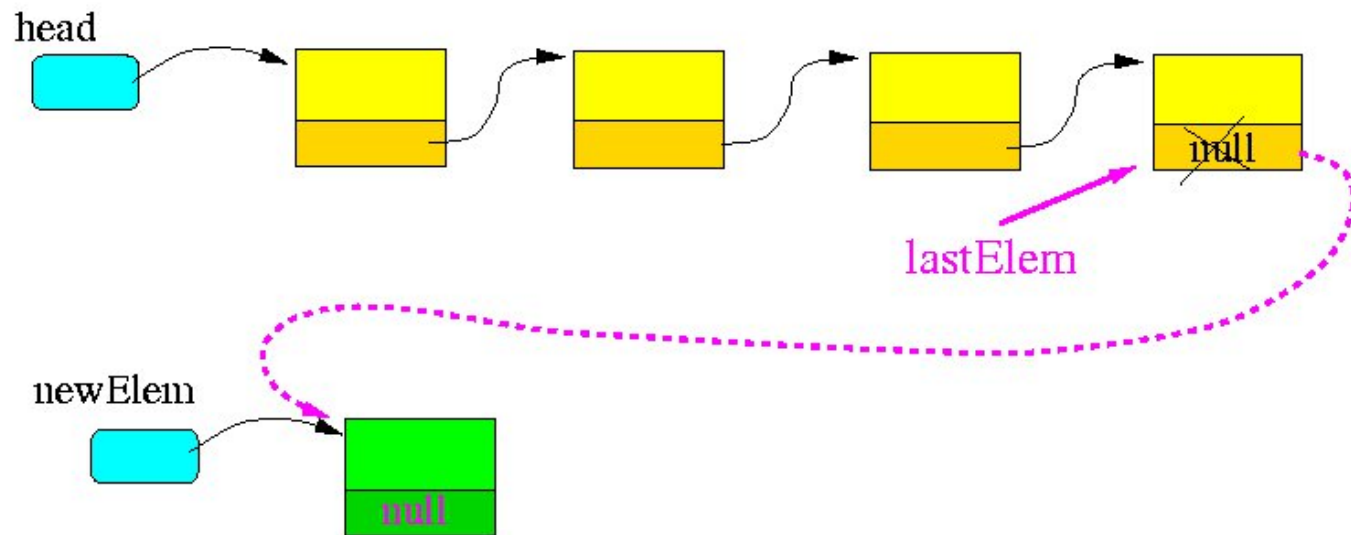
Three pointer fields are changed as follows:

1. The next pointer field of node A now points to the new node N, to which AVAIL previously pointed.
2. AVAIL now point to the second node in the free pool, to which node N previously pointed.
3. The next pointer field of node N now points to node B, to which node A previously pointed.



Insertion in the last position

What you need to do to insert at tail:



Inserting at middle or last

Step 1. OVERFLOW, IF(AVAIL = NULL)
write : OVERFLOW and EXIT.

Step 2. Create a new node NODE=AVAIL
and assign the address to any variable say AVAIL_PTR .

Step 3. ASSIGN NODE[DATA] = ITEM

Step 4. IF PREVIOUS_NODE[LINK]=NULL
Set NODE[LINK]= NULL and
ELSE
Set NODE[LINK]= PREVIOUS_NODE[LINK]

Step 5. ASSIGN PREVIOUS_NODE[LINK]=AVAIL_PTR

Step 6. EXIT



Question?